

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Software Engineering Course Code: CSPC-514 Examination: 1st Periodical Examination

Scheme: NEP Branch: CSE (AI & ML) Semester: 5th

Question Paper for 1st Periodical Examination (October 2025) (Session July-Dec 2025)

Time: One and a half hours

Max. Marks: 30

Note: Attempt all questions.

Course Outcomes:

CO1. Understand software engineering process life cycle and compare various software development models.

CO2. Elicit, analyse and specify software requirements, design and implementation of the project.

CO3. Formulate a testing strategy for a software system.

CO4. Maintain software, applying reengineering and software reuse techniques.

CO5. Understand and use agile software development concepts.

Q.1 Do the following:

(1x5=5)

- (a) List pitfalls of waterfall model?
- (b) List properties of good SRS documents.
- (c) What is a design pattern?
- (d) What are the advantages of the evolutionary process model?
- (e) What is requirement validation vs. requirement verification?

Q.2 Differentiate between functional and non-functional requirements.

(5)

Q.3 What are UML diagrams? List different types and their uses.

(5)

Q.4 What are design principles in software engineering?

(5)

Q.5 Given a project titled "Online GPS Based Vehicle Tracking System", do the following:

(10)

1. Which SDLC process model would you adopt and why?
2. Identify various stakeholders involved in it.
3. Identify functional and non-functional requirements.
4. List the various tools used in information gathering and why so?
5. Model the system as per requirement: Design use case diagram, activity diagram.
6. Design DFD for the above.

CO: Course Outcome KL: Knowledge Level: L1 - Remember, L2 - Understand, L3 - Apply, L4 - Analyze, L5 - Evaluate, L6 - Create

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